1.Explain the Structure of HTML with Example

Introduction

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. An HTML document consists of different elements and tags arranged in a specific structure.

Basic Structure of HTML
```
<!DOCTYPE html>
<html>
<head>
  <style>
    .blur {
      filter: blur(5px);
    }

    .bright {
      filter: brightness(150%);
    }

    .gray {
      filter: grayscale(100%);
    }
  </style>
</head>

<body>

  <img src="image.jpg" class="blur">
  <img src="image.jpg" class="bright">
  <img src="image.jpg" class="gray">

</body>
</html>
```

The basic structure of an HTML document is:

```
<!DOCTYPE html>
<html>
<head>
    <title>My Web Page</title>
</head>
<body>
    <h1>Welcome to My Website</h1>
    <p>This is my first web page.</p>
</body>
</html>
```

Explanation of Each Part

1. <!DOCTYPE html>

- It declares that the document is written in **HTML5**.
- It helps the browser understand which version of HTML is being used.

2. <html>

- It is the **root element** of an HTML document.
- All other HTML elements are written inside this tag.

3. <head>

- It contains information about the webpage that is generally not displayed directly on the page.
- It can contain <title>, CSS links, metadata, and scripts.

4. <title>

- It defines the title of the webpage.
- The title appears on the **browser tab**.

5. <body>

- It contains the **visible content** of the webpage.
- Headings, paragraphs, images, links, tables, forms, etc. are placed inside the body.

6. <h1>

- It defines the main or largest heading of the webpage.

7. <p>

- It defines a paragraph of text.

Example

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Page</title>
</head>
```

```html
<body>
    <h1>My College</h1>
    <p>Welcome to my college webpage.</p>
    <h2>About Me</h2>
    <p>My name is Husen. I am learning web development.</p>
</body>
</html>
```

Output

My College
Welcome to my college webpage.

About Me
My name is Husen. I am learning web development.

2.Define CSS. Explain the Types of CSS Used in Web Design

Definition of CSS

CSS (Cascading Style Sheets) is a stylesheet language used to **style and design HTML web pages**. It is used to control the appearance, layout, colors, fonts, spacing, borders, and responsiveness of a webpage.

CSS helps to **separate the content of a webpage from its presentation**, making websites easier to design and maintain.

Types of CSS

There are **three main types of CSS** used in web design:

1. **Inline CSS**
2. **Internal CSS**
3. **External CSS**

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1. Inline CSS

Inline CSS is written directly inside an HTML element using the style attribute.

Example:

<p style="color: blue; font-size: 20px;">

Welcome to my website

</p>

Advantages:

- Easy to apply to a single element.
- Useful for quick and specific styling.

Disadvantages:

- Difficult to maintain for large websites.
- Repeated styles make HTML code lengthy.

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2. Internal CSS

Internal CSS is written inside the <style> tag within the <head> section of an HTML document.

Example:

```
<!DOCTYPE html>
<html>
<head>
    <style>
        h1 {
            color: red;
            text-align: center;
        }
        p {
            font-size: 18px;
        }
    </style>
</head>
<body>
    <h1>My Website</h1>
    <p>Welcome to Web Design.</p>
</body>
</html>
```

Advantages:

- Useful when styling a single webpage.
- Keeps CSS separate from individual HTML elements.

Disadvantages:

- Cannot be easily reused across multiple webpages.
- Makes the HTML file larger.

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3. External CSS

External CSS is written in a separate .css file and connected to an HTML document using the <link> tag.

HTML file:

```
<!DOCTYPE html>
<html>
```

```
<head>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <h1>My Website</h1>
    <p>Welcome to Web Design.</p>
</body>
</html>
```

CSS file (style.css):

```
h1 {
    color: green;
    text-align: center;
}
p {
    font-size: 18px;
}
```

Advantages:

- Can be reused across multiple webpages.
- Makes websites easier to maintain.
- Keeps HTML and CSS separate.
- Suitable for large websites.

Disadvantage:

- Requires an additional CSS file.
- If the CSS file is unavailable, the styling may not be applied.

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3. Explain Types of Hyperlinks in HTML with Example

Introduction

A **hyperlink** is a clickable link that allows a user to move from one webpage or resource to another. In HTML, hyperlinks are created using the **<a> (anchor) tag**.

Basic Syntax

Link Text

Here:

- <a> → Defines a hyperlink.
- href → Specifies the destination of the link.
- Link Text → The clickable text displayed on the webpage.

Types of Hyperlinks in HTML

There are mainly **four types of hyperlinks**:

1. External Hyperlink

An **external hyperlink** connects a webpage to a page on another website or domain.

Example:

Visit Google

When the user clicks **Visit Google**, the browser opens Google's website.

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2. Internal Hyperlink

An **internal hyperlink** connects one page of a website to another page within the same website.

Example:

About Us

Here, clicking **About Us** opens the about.html page.

Another example:

Contact Us

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3. Email Hyperlink

An **email hyperlink** allows the user to open their default email application and send an email.

It uses the mailto: scheme.

Example:

Send Email

When the user clicks **Send Email**, the default email application opens with the email address filled in.

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4. Bookmark/Anchor Hyperlink

A **bookmark hyperlink** is used to move to a specific section of the **same webpage**. It is useful for long webpages.

Example:

Go to About Section

<h2 id="about">About Us</h2>

<p>This is the about section of the webpage.</p>

When the user clicks **Go to About Section**, the browser moves directly to the section having id="about".

Example of All Types

```
<!DOCTYPE html>
<html>
<head>
    <title>Types of Hyperlinks</title>
</head>
<body>

  <!-- External Link -->

  <a href="https://www.google.com">Visit Google</a>

    <br><br>
    <!-- Internal Link -->
    <a href="about.html">About Us</a>
    <br><br>


  <!-- Email Link -->

    <a href="mailto:example@gmail.com">Send Email</a>
    <br><br>

  <!-- Bookmark Link -->

  <a href="#contact">Go to Contact Section</a>

    <h2 id="contact">Contact Us</h2>
    <p>Welcome to the Contact section.</p>
</body>
</html>
```

Advantages of Hyperlinks

- Helps users navigate between webpages.
- Provides quick access to external websites.
- Allows navigation within a long webpage.
- Makes websites more interactive and user-friendly.
- Can provide direct access to email communication.

4. Explain How Text and Attributes Can Be Changed Dynamically — 10 Marks

Introduction

In web development, **JavaScript** can be used to change the content and properties of HTML elements dynamically without reloading the webpage.

- **Text** can be changed using properties such as textContent and innerHTML.
- **Attributes** can be changed using methods such as setAttribute() and removeAttribute().

This is commonly done using the **DOM (Document Object Model)**.

1. Changing Text Dynamically

JavaScript provides different ways to change the text of an HTML element.

Using textContent

The textContent property changes the text inside an HTML element.

Example:

```
<p id="message">Hello World</p>

<script>
    document.getElementById("message").textContent =
        "Welcome to JavaScript!";
</script>
```

Output:

Before:

Hello World

After:

Welcome to JavaScript!

Using innerHTML

innerHTML can change the content inside an HTML element. It can also insert HTML tags.

Example:

```
<p id="demo">Old Text</p>

<script>
    document.getElementById("demo").innerHTML =
        "<b>New Text</b>";
</script>
```

Here, the text is changed and displayed in **bold**.

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2. Changing Attributes Dynamically

HTML elements have attributes such as src, href, class, id, and title. JavaScript can change these attributes while the webpage is running.

Using setAttribute()

The setAttribute() method is used to add or change an attribute.

Example:

```
<img id="photo" src="old.jpg">

<script>
    document.getElementById("photo")
        .setAttribute("src", "new.jpg");
```

```
</script>
```

Here, the src attribute is changed from old.jpg to new.jpg.

Changing Attributes Directly

Some attributes can also be changed directly using JavaScript.

Example:

```
<a id="link" href="https://example.com">
    Visit Website
</a>

<script>
    document.getElementById("link").href =
        "https://google.com";
</script>
```

Here, the href attribute is dynamically changed.

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3. Removing an Attribute

The removeAttribute() method removes an attribute from an HTML element.

Example:

```
<input id="name" type="text" disabled>

<script>
    document.getElementById("name")
        .removeAttribute("disabled");
</script>
```

After executing the JavaScript, the disabled attribute is removed and the input becomes enabled.

Complete Example

```
<!DOCTYPE html>
<html>
<head>
    <title>Dynamic Changes</title>
</head>

<body>
    <h1 id="heading">Hello Student</h1>
    <img id="image" src="old.jpg" width="200">
    <button onclick="changeContent()">
        Change
    </button>
    <script>
        function changeContent() {
            // Change text
            document.getElementById("heading").textContent =
                "Welcome to JavaScript";
            // Change attribute
            document.getElementById("image")
                .setAttribute("src", "new.jpg");
        }
```

```
    </script>
</body>
</html>
```

When the **Change** button is clicked:

1. The heading text changes.
2. The image's src attribute changes.
3.

5. What is a CSS Filter? Give Three Examples of Filter Functions — 10 Marks

Definition

A **CSS filter** is a CSS property used to apply **visual effects to HTML elements**, especially images. Filters can change the appearance of an element by modifying its brightness, contrast, color, blur, and other visual properties.

The CSS filter property is used to apply these effects.

Syntax

```
selector {
    filter: function(value);
}
```

Three Examples of CSS Filter Functions

1. blur()

The blur() function makes an element appear **blurred**.

Example:

```
img {
    filter: blur(5px);
}
```

Here, 5px determines the amount of blur.

---

2. brightness()

The brightness() function increases or decreases the **brightness** of an element.

Example:

```
img {
    filter: brightness(150%);
}
```

Here, 150% makes the image brighter than its original appearance.

- 100% → Normal brightness
- Less than 100% → Darker

- More than 100% → Brighter

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3. grayscale()

The grayscale() function converts an element into **shades of gray**.

Example:

```
img {
    filter: grayscale(100%);
}
```

Here, 100% makes the image completely grayscale.

Complete Example

```
<!DOCTYPE html>
<html>
<head>
    <style>
        .blur {
            filter: blur(5px);
        }

        .bright {
            filter: brightness(150%);
        }

        .gray {
            filter: grayscale(100%);
        }
    </style>
</head>

<body>

    <img src="image.jpg" class="blur">
    <img src="image.jpg" class="bright">
    <img src="image.jpg" class="gray">

</body>
</html>
```

Other CSS Filter Functions

Some other commonly used filter functions are:

- contrast() — adjusts contrast
- sepia() — gives a brownish/old-photo effect
- opacity() — changes transparency
- saturate() — adjusts color intensity
- invert() — reverses the colors
- hue-rotate() — changes the color hue